CAMPANA POINTS ON QUADRICS WITH HYPERPLANE BOUNDARY

DASHENG WEI

Abstract. Let $Q \subset \mathbf{P}_k^{d+1}$ be a smooth quadric of dimension $d \geq 2$ over a number field, and let $H_i = Q \cap \Pi_i$ be hyperplane sections, for $1 \leq i \leq d$. For every integer $m \geq 2$, we prove Campana strong approximation for the pair $\big(Q, (1-1/m)(H_1 + \cdots + H_d)_{\mathrm{red}}\big)$.

1. Introduction

Let k be a number field. We study the approximation of rational points on smooth quadrics, subject to lower bounds on the positive intersection multiplicities with a hyperplane boundary.

Quadrics satisfy classical local–global principles. The Hasse–Minkowski theorem gives the Hasse principle for projective quadrics, and a smooth quadric with a rational point is k-rational and satisfies weak approximation [Har04, Theorem 2.2.1]. For affine quadrics, strong approximation requires a noncompactness condition at an omitted place, and in dimension two a Brauer–Manin condition may be needed [CTX09, Theorem 8.3 and Remark 8.4]. For rational points on intersections of two quadrics, see [CTSSD87a, CTSSD87b, SS91, Wit07, HB18, CT24].

In the Campana setting, Nakahara and Streeter established approximation results for projective spaces and for the projective line with a boundary of degree two [NS24]; the log Fano case on $\mathbf{P}^1$ is treated in [Wei26b, Theorem 1.1]. Mitankin, Nakahara and Streeter proved Campana strong approximation for a smooth quadric whose boundary is a single hyperplane section [MNS24, Theorem 1.4]. In this paper, we treat the case where the boundary is the reduced union of hyperplane sections.

For $d \geq 2$, let $Q \subset \mathbf{P}_k^{d+1}$ be a smooth quadric, let $\Pi_1, \ldots, \Pi_d$ be hyperplanes defined over k, and put

$$H_i = Q \cap \Pi_i, \qquad D = H_1 + \cdots + H_d, \qquad \Delta_m = \left(1 - \frac{1}{m}\right) D_{\mathrm{red}}. \tag{1.1}$$

Each distinct prime component of D occurs in D_{red} with multiplicity one and has weight m; the hyperplanes need not be distinct. The common intersection $E = Q \cap \Pi_1 \cap \cdots \cap \Pi_d$ is understood scheme-theoretically and may have positive dimension. For $d \geq 2$, every hyperplane section is geometrically reduced, since its quadratic form has rank at least d. For

2020 *Mathematics Subject Classification.* Primary 14G05; Secondary 11G35, 14G12.

Key words and phrases. Campana points, quadrics, hyperplane sections, strong approximation, conic fibrations, norm equations.

$d \geq 3$ it is geometrically integral. On a surface, distinct sections may share a line; that line is counted only once in D_{red}. The weight is imposed separately on each prime component and each local component. Every effective divisor on a smooth quadric of dimension at least two is nef. Hence

$$-(K_Q + \Delta_m) \sim \frac{d}{m}\mathcal{O}_Q(1) + \left(1 - \frac{1}{m}\right)(D - D_{\mathrm{red}})$$

is ample.

When E is finite, the defining linear forms are linearly independent. Then $\Lambda = \Pi_1 \cap \cdots \cap \Pi_d \simeq \mathbf{P}^1_k$ is not contained in Q, D is reduced, and E has length two. It is either reduced or of the form $2P$ for a point $P \in Q(k)$.

Theorem 1.1. *Let $Q \subset \mathbf{P}^{d+1}_k$ be a smooth quadric of dimension $d \geq 2$, and let $\Pi_1, \ldots, \Pi_d$ be hyperplanes defined over k. Put $D = \sum_{i=1}^{d}(Q \cap \Pi_i)$ and $\Delta_m = (1 - 1/m)D_{\mathrm{red}}$. Then, for every integer $m \geq 2$, the pair (Q, Δ_m) satisfies Campana strong approximation:*

$$\overline{(Q \setminus D)(k)}^{\mathrm{C}} = (Q, \Delta_m)(\mathbf{A}_k). \tag{1.2}$$

Here the closure is taken in the restricted-product Campana topology of Definition 2.1.

Corollary 1.2. *Under the hypotheses of Theorem 1.1, fix a finite $S \supseteq \Omega^{\infty}_k$ and a proper flat integral $\mathcal{O}_S$-model $\mathscr{Q}$ of (Q, Δ_m). The strict S-Campana points of this model are dense in*

$$\mathcal{A}_S(\mathscr{Q}, \Delta_m) = \prod_{v \in S} Q(k_v) \times \prod_{v \notin S} \mathcal{C}_v(\mathscr{Q}, \Delta_m) \tag{1.3}$$

with its product topology, where $\mathcal{C}_v$ is defined using this fixed model. Thus every finite collection of local approximations can be achieved while imposing the corresponding Campana inequalities at all places outside the original S.

Proof. Apply Theorem 1.1 and Proposition 2.3. □

Corollary 1.3. *Under the hypotheses of Theorem 1.1, write $D_{\mathrm{red}} = \sum_{\alpha \in I} D_\alpha$ for its distinct prime components. Let $n_\alpha \geq 2$ be arbitrary integers and put*

$$\Delta_{\boldsymbol{n}} = \sum_{\alpha \in I} \left(1 - \frac{1}{n_\alpha}\right) D_\alpha. \tag{1.4}$$

Then $(Q, \Delta_{\boldsymbol{n}})$ satisfies Campana strong approximation. The fixed-model conclusion of Corollary 1.2 also holds with these individual weights.

Proof. Put $N = \max_{\alpha \in I} n_\alpha$. Theorem 1.1 applies to $(Q, (1 - 1/N)D_{\mathrm{red}})$, and the monotonicity of Campana strong approximation [Wei26a, Lemma 2.5(1)] gives the assertion. □

We use approximation on $\mathbf{P}^1$ with a reduced boundary of degree at most two, recalled in Proposition 2.6. First suppose that $E(k) \neq \varnothing$, without assuming that E is finite, and put $L = \Pi_1 \cap \cdots \cap \Pi_d$. Choose a line $\ell \subset L$ through a rational point of E. If $\ell \subset Q$, its span with a general rational point cuts out ℓ and a residual line, with a one-point boundary. Otherwise

it cuts out a smooth conic with reduced boundary of degree at most two. Proposition 3.2 treats all these cases.

We may therefore assume $E(k) = \varnothing$; then E is automatically smooth, as explained in Remark 3.3. If E is finite, it is a closed point of degree two. For surfaces, we use the conic pencil through $H_1 \cap H_2$. The Campana formal lemma of [MNS24, Theorem 3.17], combined with the descent argument in [HWW22, Lemma 4.2], produces local Campana points admitting lifts to a twisted norm auxiliary variety. Following this fibration method, strong approximation on the auxiliary variety selects a smooth fiber with local points forming a tuple in the prescribed Campana-open set; see Lemma 4.4. Campana strong approximation on $\mathbf{P}^1$ then give the desired rational point. For $d > 2$, the common plane conic $B = Q \cap \Pi_2 \cap \cdots \cap \Pi_d$ is automatically reduced. Passing to a suitably chosen smooth quadric surface through B reduces the approximation problem to the surface case with two conic boundary sections.

It remains to treat positive-dimensional E with $E(k) = \varnothing$. The quadratic form on L is then nondegenerate. Choose a plane $L' \subset L$ with smooth conic $B = Q \cap L'$. For general R, the section $\Sigma_R = Q \cap \langle L', R\rangle$ is a smooth quadric surface, and every H_i restricts to B. The one-boundary surface case in Corollary 4.5 completes this reduction.

After the preliminaries in Section 2, Section 3 treats all E with $E(k) \neq \varnothing$. The remaining surface case is proved in Section 4. Section 5 gives the remaining reductions to surfaces, for finite and positive-dimensional E, and completes the proof.

2. Campana approximation and preliminary results

Write Ω_k for the set of places of k and Ω_k^∞ for the archimedean places. For a finite place v, write $\mathcal{O}_v$, $\mathfrak{m}_v$ and $\mathbf{F}_v$ for its valuation ring, maximal ideal and residue field. We normalize ord_v to have image $\mathbf{Z}$. For finite $S \supseteq \Omega_k^\infty$, let $\mathcal{O}_S$ be the ring of S-integers.

2.1. **Campana topology.** Let X be a smooth geometrically integral k-variety, not necessarily proper. Let $D = \sum_i D_i$ be a reduced divisor with distinct prime components defined over k, and put

$$\Delta = \sum_i \left(1 - \frac{1}{m_i}\right) D_i, \qquad m_i \in \mathbf{Z}_{\geq 2}.$$

All weights on X are finite. Fix a finite $S \supseteq \Omega_k^\infty$ and a flat separated integral $\mathcal{O}_S$-model $\mathscr{X}$ of finite type, with generic fiber X. Let $\mathscr{D}_i$ be the reduced closure of D_i.

For $v \notin S$, write $D_i \times_k k_v = \bigcup_\lambda D_{i,\lambda}$ for the local irreducible components, and let $\mathscr{D}_{i,\lambda}$ be their reduced closures in $\mathscr{X}_{\mathcal{O}_v}$. For $P \in \mathscr{X}(\mathcal{O}_v) \subset X(k_v)$, let $\widetilde{P} : \operatorname{Spec} \mathcal{O}_v \to \mathscr{X}_{\mathcal{O}_v}$ be its section. Separatedness makes this extension unique.

For any closed subscheme $\mathscr{Z} \subset \mathscr{X}_{\mathcal{O}_v}$, let $\mathcal{I}_{\mathscr{Z}} \subset \mathcal{O}_{\mathscr{X}_{\mathcal{O}_v}}$ be its ideal sheaf. Define its pulled-back ideal along $\widetilde{P}$ by

$$J_{\mathscr{Z},P} := \operatorname{im}\big(\widetilde{P}^*\mathcal{I}_{\mathscr{Z}} \longrightarrow \mathcal{O}_v\big) = \widetilde{P}^{-1}\mathcal{I}_{\mathscr{Z}} \cdot \mathcal{O}_v \subseteq \mathcal{O}_v.$$

If $f_1, \dots, f_r$ generate $\mathcal{I}_{\mathscr{Z}}$ near the reduction of P, then $J_{\mathscr{Z},P} = (f_1(P), \dots, f_r(P))$. In particular, write $J_{i,\lambda,P} := J_{\mathscr{D}_{i,\lambda},P}$. Since $\mathcal{O}_v$ is a discrete valuation ring, a nonzero $J_{i,\lambda,P}$ equals $\mathfrak{m}_v^n$ for a unique integer $n \geq 0$. Put

$$n_v(\mathscr{D}_{i,\lambda}, P) = \begin{cases} \operatorname{length}_{\mathcal{O}_v}(\mathcal{O}_v/J_{i,\lambda,P}), & J_{i,\lambda,P} \neq 0, \\ \infty, & J_{i,\lambda,P} = 0, \end{cases} \tag{2.1}$$

Thus a local equation f gives $n_v = \operatorname{ord}_v(f(P))$ whenever that equation generates the pulled-back ideal.

Define

$$\begin{aligned} &\mathcal{C}_v(\mathscr{X}, \Delta) = \{P \in \mathscr{X}(\mathcal{O}_v) : n_v(\mathscr{D}_{i,\lambda}, P) \in \{0\} \cup \mathbf{Z}_{\geq m_i} \cup \{\infty\} \text{ for every } i, \lambda\}, && v \notin S, \\ &\mathcal{C}_v(\mathscr{X}, \Delta) = X(k_v), && v \in S, \\ &\mathcal{C}_S(\mathscr{X}, \Delta) = \{P \in (X \setminus D)(k) : P \in \mathcal{C}_v(\mathscr{X}, \Delta) \text{ for every } v \notin S\}. \end{aligned}$$

A local point $P \in \mathcal{C}_v(\mathscr{X}, \Delta)$ is called a *strict local Campana point* if $P \in (X \setminus D)(k_v)$. A rational point $P \in (X \setminus D)(k)$ is called a *strict S-Campana point* for the fixed model if it extends to an $\mathcal{O}_v$-point of $\mathscr{X}$ for every $v \notin S$ and satisfies

$$n_v(\mathscr{D}_{i,\lambda}, P) = 0 \quad \text{or} \quad n_v(\mathscr{D}_{i,\lambda}, P) \geq m_i \qquad \text{for every } v \notin S,\ i, \lambda.$$

Thus $\mathcal{C}_S(\mathscr{X}, \Delta)$ is precisely the set of strict S-Campana points. Strictness requires that the point avoid the boundary on the generic fiber; reduction onto the boundary is permitted whenever each positive intersection multiplicity is at least the corresponding bound m_i. The integrality requirement outside S is automatic when $\mathscr{X}$ is proper. We use the local-component convention of [NS24, Definitions 2.2–2.4]. Boundary-valued local points are allowed; global approximating points are required to be strict. For equal weights $m_i = m$, this gives $\Delta = (1 - 1/m)D$.

For the fixed S and model, write

$$\mathcal{A}_S(\mathscr{X}, \Delta) = \prod_v \mathcal{C}_v(\mathscr{X}, \Delta)$$

with the product topology.

Definition 2.1 (Campana topology)**.** The *Campana adelic space* is

$$(X, \Delta)(\mathbf{A}_k) = \prod_v{}' (X(k_v), \mathcal{C}_v(\mathscr{X}, \Delta)) = \bigcup_{T \supseteq S \text{ finite}} \mathcal{A}_T(\mathscr{X}, \Delta). \tag{2.2}$$

Its points satisfy the local Campana conditions at almost every place. We equip it with the restricted-product topology, which we call the *Campana topology*. A basis consists of

$$\prod_{v \in T} W_v \times \prod_{v \notin T} \mathcal{C}_v(\mathscr{X}, \Delta), \qquad T \supseteq S \text{ finite}, \tag{2.3}$$

where $W_v \subset X(k_v)$ is open in the usual local analytic topology. We reserve $\mathcal{A}_S(\mathscr{X},\Delta)$ for the fixed-model space and use $(X,\Delta)(\mathbf{A}_k)$ for the model-independent Campana adelic space.

Following [Wei26a, Definition 2.1], the pair (X,Δ) satisfies *Campana strong approximation* if $(X\setminus D)(k)$ is dense in $(X,\Delta)(\mathbf{A}_k)$. Put

$$(X,\Delta)(\mathbf{A}_k)^{\mathrm{Br}} = (X,\Delta)(\mathbf{A}_k)\cap X(\mathbf{A}_k)^{\mathrm{Br}(X)},$$

where the second factor is the usual Brauer–Manin set. Campana strong approximation with the Brauer–Manin obstruction means

$$\overline{(X\setminus D)(k)}^{\mathrm{C}} = (X,\Delta)(\mathbf{A}_k)^{\mathrm{Br}}.$$

The fixed model satisfies *Campana weak approximation* if $\mathcal{C}_S(\mathscr{X},\Delta)$ is dense in $\mathcal{A}_S(\mathscr{X},\Delta)$ with its product topology.

The topology in Definition 2.1 uses local Campana sets as reference sets, as in [Wei26a, Section 2.2]. For a related construction of adelic M-points, see [Moe24, Definition 4.1]. It is independent of the model and the reference set S, since two models and their boundary closures agree outside finitely many places. Since $X\setminus D$ is geometrically integral, Lang–Weil and Hensel's lemma give $\mathcal{C}_v(\mathscr{X},\Delta)\neq\varnothing$ for all but finitely many v. Thus the finite reference set S in (2.2) may be chosen so that these sets are nonempty for every $v\notin S$.

We recall [Wei26a, Lemma 2.3 and Proposition 2.4(1)]; the same arguments apply to the separated models above.

Lemma 2.2. *For finite weights, $\mathcal{C}_v(\mathscr{X},\Delta)$ is open and closed in $X(k_v)$. Its intersection with every dense Zariski-open subset of X is dense in $\mathcal{C}_v(\mathscr{X},\Delta)$.*

Consequently, $\mathcal{A}_S(\mathscr{X},\Delta)$ is open in $(X,\Delta)(\mathbf{A}_k)$ and carries its product topology.

Proposition 2.3. *Campana strong approximation for (X,Δ) is equivalent to Campana weak approximation for every finite $S\supseteq\Omega_k^\infty$ and every flat separated integral model of finite type.*

2.2. **Orbifold morphisms and continuity.**

Definition 2.4 (Orbifold morphism)**.** Let (Y,Δ_Y) and (X,Δ_X) be smooth geometrically integral k-varieties with finite-weight Campana boundaries. Each geometric boundary component inherits its weight; write $m_Y(F)=1$ for a geometric prime divisor F outside the boundary of $Y_{\overline{k}}$. Following [Cam11, Definition 2.3], a k-morphism $f:Y\to X$ is a *morphism of Campana orbifolds* if $f(Y)\not\subseteq \operatorname{Supp}\Delta_X$ and, for every geometric boundary component D of $X_{\overline{k}}$,

$$f_{\overline{k}}^{*}D=\sum_F e_{F,D}F \quad (e_{F,D}>0) \quad\Longrightarrow\quad e_{F,D}\,m_Y(F)\geq m_X(D). \tag{2.4}$$

Here $f_{\overline{k}}^{*}D$ is the effective Cartier pullback of D: if D has local equation $g=0$, its pullback has equation $g\circ f_{\overline{k}}=0$. It is well-defined because $X_{\overline{k}}$ is smooth and $f_{\overline{k}}(Y_{\overline{k}})$ is not contained in D. The sum runs over the prime components of its support, and $e_{F,D}$ is the order of vanishing along F. For an inclusion $j:Y\hookrightarrow X$, this is the scheme-theoretic divisor restriction $D|_Y$,

with its multiplicities. The inequalities are checked on geometric components, in accordance with our local-component convention. Dominance and smoothness of the boundary are not required.

The following elementary functoriality statement uses the Campana topology of Definition 2.1 on both adelic spaces. We include a proof adapted to our local-component convention.

Proposition 2.5 (Continuity)**.** *An orbifold morphism $f : (Y, \Delta_Y) \to (X, \Delta_X)$ induces a continuous map*

$$f_{\mathbf{A}} : (Y, \Delta_Y)(\mathbf{A}_k) \longrightarrow (X, \Delta_X)(\mathbf{A}_k).$$

More precisely, for any fixed models $\mathscr{Y}$ of Y and $\mathscr{X}$ of X, there is a finite set $T \supseteq \Omega_k^\infty$, containing the places removed from both models, such that

$$f\big(\mathcal{C}_v(\mathscr{Y}, \Delta_Y)\big) \subseteq \mathcal{C}_v(\mathscr{X}, \Delta_X) \qquad (v \notin T). \tag{2.5}$$

In particular, the inverse image of a Campana-open set is open.

Proof. Choose a finite Galois extension L/k over which all geometric components of the boundaries and of the divisors $f_{\overline{k}}^* D$ are defined. Choose a finite T containing the places removed from both models and the ramified places of L/k, so that f extends to the fixed models and these finitely many divisor identities hold for their flat Cartier closures over the integral closure of $\mathcal{O}_T$ in L. For $v \notin T$ and $w \mid v$, the k_v-irreducible components correspond to $\mathrm{Gal}(L_w/k_v)$-orbits of the geometric components. Summing the identities over these orbits and descending along $\mathcal{O}_{L_w}/\mathcal{O}_v$ gives the identities for the local component closures. Thus they hold for every local component outside the same finite set T. For a local irreducible component D_λ of the boundary on X, write $f^* D_\lambda = \sum_\alpha e_{\alpha,\lambda} F_\alpha$ ($e_{\alpha,\lambda} > 0$), summing over the prime components of its support. The geometric inequalities imply $e_{\alpha,\lambda} m_Y(F_\alpha) \geq m_X(D_\lambda)$, since each local coefficient is a sum of the relevant geometric coefficients. The pullback identity for intersection multiplicities [Moe24, Proposition 3.27] gives, for $P \in \mathcal{C}_v(\mathscr{Y}, \Delta_Y)$ and $v \notin T$,

$$n_v(\mathscr{D}_\lambda, f(P)) = \sum_\alpha e_{\alpha,\lambda}\, n_v(\mathscr{F}_\alpha, P). \tag{2.6}$$

Every positive $n_v(\mathscr{F}_\alpha, P)$ is at least $m_Y(F_\alpha)$, with weight one assigned to unmarked components. Hence every positive $n_v(\mathscr{D}_\lambda, f(P))$ is at least $m_X(D_\lambda)$; infinite intersection multiplicities are allowed. This proves the inclusion (2.5). The inclusion shows that $f_{\mathbf{A}}$ is well-defined. To prove continuity, let $\mathcal{N} = \prod_v N_v$ be a basic neighborhood of $f_{\mathbf{A}}((y_v))$ in $(X, \Delta_X)(\mathbf{A}_k)$. Choose a finite B containing T, the prescribed places of $\mathcal{N}$, and all places where $y_v \notin \mathcal{C}_v(\mathscr{Y}, \Delta_Y)$. At $v \in B$, local continuity gives an open neighborhood U_v of y_v with $f(U_v) \subseteq N_v$. Then

$$\prod_{v \in B} U_v \times \prod_{v \notin B} \mathcal{C}_v(\mathscr{Y}, \Delta_Y) \subseteq f_{\mathbf{A}}^{-1}(\mathcal{N})$$

is a basic neighborhood of (y_v) in $(Y, \Delta_Y)(\mathbf{A}_k)$, by the inclusion outside T. This proves continuity. □

In particular, an inclusion $f : Y \hookrightarrow X$ whose geometric boundary pullbacks are supported on the boundary of $Y_{\overline{k}}$ is an orbifold morphism whenever $e_{F,D} m_Y(F) \geq m_X(D)$ for every pullback component. Strict points on (Y, Δ_Y) map to strict points on (X, Δ_X) when $f^{-1}(\operatorname{Supp} \Delta_X) \subseteq \operatorname{Supp} \Delta_Y$. This support condition holds for the conic and surface inclusions used below.

To prove Campana strong approximation, we shall meet an arbitrary nonempty Campana-open set $\mathcal{N}$. By Lemma 2.2, we may shrink it to a basic set (2.3) with strict local factors, including any given finite set of places among its finite local conditions. For a proper variety, we use a proper reference model in these basic presentations.

2.3. **Approximation results.** We use the following result on curves.

Proposition 2.6 (Nakahara–Streeter)**.** *Let $C \simeq \mathbf{P}^1_k$, and let $E \subset C$ be a reduced divisor of degree at most two. Assign to each closed point $P \in \operatorname{Supp}(E)$ an arbitrary integer $n_P \geq 2$ and put*

$$\Delta_E = \sum_{P \in \operatorname{Supp}(E)} \left(1 - \frac{1}{n_P}\right) P.$$

Then (C, Δ_E) satisfies Campana strong approximation.

Proof. If $E = 0$, use ordinary weak approximation on $\mathbf{P}^1$. If every point of E is rational, apply [NS24, Proposition 3.5] with $n = 1$. Otherwise E is a single closed point of degree two, and [NS24, Proposition 3.15] applies. The assertion for every proper model follows from [NS24, Lemma 2.8 and Remark 3.16]; Proposition 2.3 gives Campana strong approximation. For a more general statement, see [Wei26b, Theorem 1.1]. □

We also use ordinary strong approximation on affine cones in the following case.

Theorem 2.7 (Wei–Xu–Zhu)**.** *Let $V \subset \mathbf{P}^n_k$, $n \geq 7$, be a pure geometrically integral intersection of two quadrics, not a cone, and suppose that V has finitely many singular points. Let W be its affine cone and let $Z \subset W$ be closed of codimension at least two. Then $W^{\mathrm{sm}} \setminus Z$ satisfies strong approximation off any one place of k.*

This is [WXZ25, Theorem 1.6(iii)]. In particular, no Brauer–Manin condition is required in this dimension.

Finally, smooth quadrics over number fields satisfy the Hasse principle and weak approximation; see, for example, [Har04, Theorem 2.2.1]. If an adelic approximation problem on Q is locally solvable, the Hasse principle gives $Q(k) \neq \varnothing$. Weak approximation, together with the density of a Zariski-open subset in each local locus, then supplies a rational point in any specified nonempty Zariski-open subset with finitely many prescribed local approximations.

We use the following version of the Campana formal lemma [MNS24, Theorem 3.17], a variant of Harari's formal lemma [Har94, Corollary 2.6.1]. We follow the local-component convention of Section 2.1 and the Campana topology of Definition 2.1. In the compactification version, coefficient one is assigned to the divisors at infinity (in $Y^c \setminus Y$), and finite auxiliary weights to additional omitted divisors (in $Y \setminus V$ but not in D).

Lemma 2.8 (Campana formal lemma)**.** *Let Y be smooth and geometrically integral, and let Δ have finite weights. Let $V \subset Y \setminus \operatorname{Supp}\Delta$ be a dense open subset, and let $\mathcal{B} \subset \operatorname{Br} V$ be a finite subgroup. Suppose $(p_v) \in (Y,\Delta)(\mathbf{A}_k)$ is orthogonal to $\mathcal{B} \cap \operatorname{Br} Y$. For every neighborhood $\mathcal{N}$ of (p_v) in the Campana topology, there exists*

$$(p'_v) \in (Y,\Delta)(\mathbf{A}_k) \cap V(\mathbf{A}_k)^{\mathcal{B}} \quad \textit{with} \quad (p'_v) \in \mathcal{N}.$$

Thus (p'_v) can be chosen arbitrarily close to (p_v) in the Campana topology.

3. A rational point in the common intersection

Let $Q \subset \mathbf{P}(V)$ and $H_1, \dots, H_d$ be as in Theorem 1.1, and let q be the nondegenerate quadratic form defining Q. Write

$$L = \bigcap_{i=1}^{d} \Pi_i = \mathbf{P}(W), \qquad E = Q \cap L.$$

We do not assume that E is finite.

Let $\ell \subset L$ be a k-line. If $\ell \subset Q$, then for general $R \in Q \setminus D$ the plane $\langle \ell, R\rangle$ is not contained in Q, and

$$Q \cap \langle \ell, R\rangle = \ell \cup C_R,$$

where C_R is a distinct k-line through R. Every Π_i cuts this plane along ℓ, so $H_i|_{C_R} = P_R$, where $P_R = \ell \cap C_R \in E(k)$.

If $\ell \not\subset Q$, put $Z = Q \cap \ell$. For general $R \in Q \setminus D$, the plane $\langle \ell, R\rangle$ cuts out a smooth conic C_R, and $H_i|_{C_R} = Z$ for every i. Indeed, the restriction of q to the two-dimensional space defining ℓ has rank two or one. In rank two, adjoining a vector of nonzero norm in its orthogonal complement gives a nondegenerate three-dimensional restriction. The resulting smooth conic contains a geometric point $R \notin \ell$. Thus the smoothness locus in $Q \setminus \ell$ is nonempty; it meets $Q \setminus D$ by irreducibility of Q. In rank one, $Z = 2P$ for some $P \in Q(k)$, and the conic is smooth whenever $R \notin T_P Q$. For rational R, the conic contains R and is therefore isomorphic to $\mathbf{P}^1_k$.

Lemma 3.1 (Reduction to a rational curve)**.** *Let $C \subset Q$ be a smooth plane conic or a k-line, not contained in D, and suppose $F = (D|_C)_{\mathrm{red}}$ has degree at most two. Put $\delta_C = (1-1/m)F$ and let $j : C \hookrightarrow Q$. If a Campana-open set $\mathcal{N} \subset (Q,\Delta_m)(\mathbf{A}_k)$ has nonempty inverse image*

$$\mathcal{G} = j_{\mathbf{A}}^{-1}(\mathcal{N}) \subset (C,\delta_C)(\mathbf{A}_k),$$

then $\mathcal{N}$ contains a point of $(C \setminus F)(k)$.

Proof. Each geometric boundary pullback is supported on F, so j is an orbifold morphism and $\mathcal{G}$ is open by Proposition 2.5. Its nonemptiness gives local points on C everywhere; the Hasse principle yields $C \simeq \mathbf{P}^1_k$ when C is a conic. Proposition 2.6 gives a strict rational point in $\mathcal{G}$, whose image lies in $\mathcal{N}$. □

Proposition 3.2. *Under the hypotheses of Theorem 1.1, assume that $E(k) \neq \varnothing$. Then (Q,Δ_m) satisfies Campana strong approximation for every $m \geq 2$.*

Proof. Let $\mathcal{N} \subset (Q, \Delta_m)(\mathbf{A}_k)$ be nonempty and open. Choose $A \in E(k)$ and a k-line $\ell \subset L$ through A. For general $R \in (Q \setminus D)(k)$, let $C = C_R$ be the residual line if $\ell \subset Q$, and the smooth conic $Q \cap \langle \ell, R \rangle$ otherwise. Put $F = (D|_C)_{\mathrm{red}}$. In the line case $F = \ell \cap C$ is one rational point; in the conic case $F = (Q \cap \ell)_{\mathrm{red}}$ has degree at most two.

Shrink $\mathcal{N}$ to the basic form (2.3), with local factors W_v at a finite set T, large enough that A satisfies the local conditions outside T. We choose $R \in W_v$ for all $v \in T$ so that some $P \in F(k)$ also satisfies these conditions outside T. If $\ell \not\subset Q$, take $P = A$ and use weak approximation on Q. If $\ell \subset Q$ and $d \geq 3$, use weak approximation again: $P = \ell \cap C$ lies on every boundary component, since all H_i are geometrically integral.

For $d = 2$ and $\ell \subset Q$, write $H_i = \ell + C_{s_i}, i = 1, 2$. Let $f : Q \to \mathbf{P}^1$ be the ruling whose fibres are the residual lines, and put $Z = \{s_1, s_2\}_{\mathrm{red}}$. Include in T bad-reduction places of Z and the exceptions for this ruling and its boundary fibres. Propositions 2.6 give a strict T-Campana parameter t for $(\mathbf{P}^1, (1 - 1/m)Z)$ in every $f(W_v)$, which is open since f is smooth. Weak approximation on C_t gives R in every W_v. The point $P = \ell \cap C_t$ has infinite multiplicity with ℓ; outside T, its multiplicities with the other boundary components are those of t with Z. Thus P satisfies the required local conditions.

The tuple equal to R at T and to P elsewhere belongs to $(C, (1 - 1/m)F)(\mathbf{A}_k)$ and maps into $\mathcal{N}$. Lemma 3.1 gives the proof. □

Remark 3.3. After Proposition 3.2, we may assume $E(k) = \varnothing$. Then $q|_W$ is nonzero and nondegenerate: the zero form would give $E = L$ with rational points, while a nonzero radical would give the rational vertex $\mathbf{P}(\mathrm{rad}(q|_W)) \subset E$. Thus E is smooth. If it is finite, it is a nonsplit degree-two point; if it has positive dimension, it is a smooth geometrically integral quadric.

4. Quadric surfaces

Let $Q \subset \mathbf{P}^3_k$ be a smooth quadric surface, and let Π_1, Π_2 be arbitrary hyperplanes defined over k. Put

$$H_i = Q \cap \Pi_i, \qquad E = H_1 \cap H_2, \qquad D = H_1 + H_2, \qquad \Delta_m = (1 - 1/m)D_{\mathrm{red}}.$$

The sections may coincide or share a line. A singular section is the union of two distinct geometric lines. Each distinct prime component has weight m, with the local Campana conditions imposed componentwise.

Proposition 4.1. *For any two hyperplane sections H_1, H_2 and every integer $m \geq 2$, (Q, Δ_m) satisfies Campana strong approximation.*

It suffices to prove the proposition when E is finite. Indeed, if $\Pi_1 \neq \Pi_2$ and E has positive dimension, then $E = \Pi_1 \cap \Pi_2$ is a k-line contained in Q, so Proposition 3.2 applies. If $\Pi_1 = \Pi_2$, choose an auxiliary hyperplane section F with $H_1 \cap F$ finite. Once the finite-intersection case is proved, applying it to $H_1 + F$ and then monotonicity [Wei26a, Lemma 2.5(1)] gives the result for $D_{\mathrm{red}} = H_1$.

The finite case with $E(k) \neq \varnothing$ also follows from Proposition 3.2. For the remaining argument, assume that E is a nonsplit closed point of degree two. Then D is reduced, so

$\Delta_m = (1-1/m)D$, and the vertex of any singular boundary section lies outside the other section.

4.1. **The conic pencil and its local geometry.** Let $\Lambda = \Pi_1 \cap \Pi_2$. The planes through Λ give a pencil of conics on Q with base locus E. Let $\pi : X = \mathrm{Bl}_E(Q) \to Q$. The pencil resolves to a morphism $f : X \to \mathbf{P}^1_k$. Its boundary has two vertical fibers, the strict transforms of H_1 and H_2, and a horizontal exceptional divisor of degree two over the base. Every smooth member C satisfies

$$H_1|_C = H_2|_C = E \tag{4.1}$$

as effective Cartier divisors. More precisely, write $E_{\overline{k}} = \{A, B\}$. If H_i is singular, its two geometric lines $L_{i,A}$ and $L_{i,B}$ satisfy

$$L_{i,A}|_C = A, \qquad L_{i,B}|_C = B.$$

A smooth section restricts to $A + B$. Thus the inclusion $(C, (1-1/m)E) \hookrightarrow (Q, \Delta_m)$ is an orbifold morphism, and Proposition 2.5 gives a continuous map on Campana adeles.

Write $L = k(E)$, a quadratic extension of k. The nondegenerate binary quadratic form on Λ is a scalar multiple of the norm form of L/k. Completing squares in its two coordinates gives the normal form

$$Q: \quad N_{L/k}(u) = p(t,w) = at^2 + btw + cw^2, \qquad H_1 = (t=0), \quad H_2 = (w=0), \tag{4.2}$$

where $u \in L$ represents a pair of k-coordinates. Smoothness of Q gives

$$b^2 - 4ac \neq 0. \tag{4.3}$$

The sections H_1 and H_2 are singular exactly when $c = 0$ and $a = 0$, respectively. The member of parameter $\tau = t/w$ is the conic

$$C_\tau : \quad N_{L/k}(u) = p(\tau, 1)w^2, \qquad t = \tau w. \tag{4.4}$$

Let $Z \subset \mathbf{P}^1$ be the reduced degree-two zero scheme of the homogeneous form $p(t,w)$. It indexes the two singular geometric members; over a closed point $s \in Z$, the corresponding member splits after adjoining L to $k(s)$. Define the reduced marked set

$$M = \{0, \infty\} \cup Z, \qquad r = \deg M \in \{2, 3, 4\}.$$

Here $r = 4, 3, 2$ according as neither, exactly one, or both boundary sections are singular. A singular boundary parameter is included only once.

Lemma 4.2. *Use the projective model of (4.2) outside a fixed finite set of places, and suppose that v is inert in L. For $P = [u : t : w] \in Q(k_v)$ with $tw \neq 0$ and $\tau = t/w$, one has*

$$n_v(\mathscr{H}_1, P) = \max\{\mathrm{ord}_v(\tau), 0\}, \qquad n_v(\mathscr{H}_2, P) = \max\{-\mathrm{ord}_v(\tau), 0\}. \tag{4.5}$$

Proof. We may assume that L/k is unramified at v, that the norm form has nondegenerate reduction, that all its coefficients are integral. In primitive integral homogeneous coordinates, t and w cannot both reduce to zero: the norm form over $\mathbf{F}_v$ is anisotropic since v is inert in L, so the coordinates of u would also reduce to zero. Consequently $\min\{\mathrm{ord}_v(t), \mathrm{ord}_v(w)\} = 0$. The two multiplicities are $\mathrm{ord}_v(t)$ and $\mathrm{ord}_v(w)$, which gives (4.5). □

Outside a fixed finite set of places, the points of E have separated reduction, and every boundary branch and vertex has good reduction, with each vertex still disjoint from the other section. At a place split in L, both points of E are k_v-rational. On any smooth member of the pencil, strict points sufficiently close to either point have arbitrarily large intersection multiplicities with the boundary components passing through that point, and zero intersection multiplicity with the remaining components. They therefore satisfy the componentwise Campana conditions.

4.2. **Twisted norm auxiliary varieties.** Assume that E is nonsplit, with splitting field L, and use the reduced marked set M defined above. Write

$$Z \setminus \{0, \infty\} = \coprod_j \operatorname{Spec} F_j. \tag{4.6}$$

For each F_j, choose a quadratic field A_j/F_j containing an embedded copy of L: take F_jL when L does not embed in F_j, and any quadratic extension otherwise. Assign to $s \in M$ the extensions $L_s/k(s)$ given by L/k at the boundary parameters and by A_j/F_j at the remaining singular parameters. If $r \leq 3$, all marked points are rational and $L_s = L$ for every $s \in M$. Every L_s contains L, and these fields are fixed throughout. Each L_s also contains the algebraic closure of $k(s)$ in the function field of a multiplicity-one component of $f^{-1}(s)$, as permitted in [HWW22, Theorem 4.1].

Choose an auxiliary affine coordinate in which M stays away from infinity, and write $a_s \in k(s)$ for the coordinate of s. For arbitrary $b_s \in k(s)^\times$, let $p_b : W_b \to \mathbf{P}^1$ be the standard auxiliary variety of [HWW22, Section 3.1], with equations

$$N_{L_s/k(s)}(z_s) = b_s(\lambda - a_s\mu), \qquad s \in M, \qquad p_b = [\lambda : \mu]. \tag{4.7}$$

Delete $\lambda = \mu = 0$ and the norm-singular loci specified in that construction. The base map is understood in the original pencil coordinates after undoing the auxiliary change of coordinates.

Lemma 4.3. *For every choice of the nonzero coefficients b_s, the variety W_b satisfies strong approximation off any finite place of k.*

Proof. Over $\overline{k}$, the norm equations have r blocks:

$$z_{j,+}z_{j,-} = \gamma_j(\lambda - a_j\mu), \qquad 1 \leq j \leq r, \tag{4.8}$$

with $\gamma_j \neq 0$ and distinct a_j.

Suppose first that $r = 4$. Eliminating λ, μ gives an affine cone over an intersection $V \subset \mathbf{P}^7$ of two quadrics. Its pencil has generic rank eight and exactly four rank-six members. The quadrics have no common factor, so [CTSSD87a, Lemmas 1.11 and 1.12] give pure geometric integrality and the fact that V is not a cone. The Jacobian criterion gives eight singular points, namely the two coordinate points in each block.

Each block-zero locus has dimension four in the six-dimensional cone: its base lies on one line, with three one-dimensional norm fibers. Simultaneous vanishing of the coordinates in two different blocks forces $\lambda = \mu = 0$, and this locus also has dimension four. Thus the

standard deleted norm-singular loci have codimension at least two, and W_b is the remaining open subset of the smooth cone. Theorem 2.7 applies.

If $r = 3$, relabel the singular boundary section as H_1. Then $p(t, w) = a\,t(t - \beta w)$ with $a\beta \neq 0$, and the marked parameters are $0, \infty, \beta$. In these coordinates the auxiliary equations are

$$N_{L/k}(x) = b_0\lambda, \qquad N_{L/k}(y) = b_\infty\mu, \qquad N_{L/k}(z) = b_\beta(\lambda - \beta\mu).$$

Eliminating λ, μ gives the nondegenerate rank-six quadric cone

$$N_{L/k}(z) = \frac{b_\beta}{b_0}N_{L/k}(x) - \frac{\beta b_\beta}{b_\infty}N_{L/k}(y).$$

Its projectivization is a smooth quadric of dimension four. The standard deleted loci have codimension at least two, so [HWW22, Lemma 6.4] gives strong approximation.

If $r = 2$, then $p(t, w) = b\,tw$ with $b \neq 0$. The equations are $N_{L/k}(x) = b_0\lambda$ and $N_{L/k}(y) = b_\infty\mu$. Eliminating λ, μ identifies W_b with an open subset of $\mathbf{A}^4_k$ obtained by deleting $x = 0$, $y = 0$, and $N_{L/k}(x) = N_{L/k}(y) = 0$. These loci have codimension at least two, and strong approximation follows from [Wei21, Lemma 2.1]. No rational baseline point or special choice of the b_s is required in any of the three cases. □

4.3. **Choosing a fiber and treating new places.**

Lemma 4.4 (Choice of a fiber)**.** *Assume that E is nonsplit. For every nonempty open subset $\mathcal{N} \subset (Q, \Delta_m)(\mathbf{A}_k)$, there is $\tau \in (\mathbf{P}^1 \setminus M)(k)$ such that, for $j_\tau : C_\tau \hookrightarrow Q$ and $\delta_\tau = (1 - 1/m)E$, the set*

$$j^{-1}_{\tau,\mathbf{A}}(\mathcal{N}) \subset (C_\tau, \delta_\tau)(\mathbf{A}_k)$$

is nonempty.

Proof. Choose $(x_v) \in \mathcal{N}$. The Hasse principle gives $Q(k) \neq \varnothing$, so Q is k-rational and $\operatorname{Br} Q = \operatorname{Br} k$. Apply the descent argument in [HWW22, Lemma 4.2] to $f : X = \operatorname{Bl}_E(Q) \to \mathbf{P}^1_k$, with $A = 0$ and $K_s = L_s$, using Lemma 2.8 for its formal-lemma step. Namely, remove M and the auxiliary infinity point from the base, and remove the exceptional divisor on X, obtaining an open subset $Q^\circ \subset Q$. The finite group of cup products used in descent meets $\operatorname{Br} Q$ only in constants, so orthogonality is automatic. We obtain $(x'_v) \in \mathcal{N} \cap Q^\circ(\mathbf{A}_k)$, coefficients b_s, and lifts $(z_v) \in W_b(\mathbf{A}_k)$ with $p_b(z_v) = f(x'_v)$. The auxiliary infinity fiber receives a finite weight in the formal lemma and adds no norm equation.

Shrink $\mathcal{N}$ about (x'_v) to a basic open set. Choose a finite B containing the local restrictions of $\mathcal{N}$, all geometric and reduction exceptions, the exceptional places of Lemma 4.2, the nonunit places of the defining and twisting coefficients, and the places where the lifts are not integral. At B, choose neighborhoods of x'_v small enough to retain membership in $\mathcal{N}$. Choose a finite place $v_0 \notin B$ splitting completely in the normal closures of all L_s. Lemma 4.3 supplies a rational point on W_b, arbitrarily close to z_v at B and integral outside $B \cup \{v_0\}$, whose parameter τ avoids M. Smooth local sections give points on C_τ in the chosen neighborhoods at B.

If $v \notin B \cup \{v_0\}$ and τ reduces to a marked point s, the induced place of $k(s)$ has degree one over v, since the marked points have étale reduction and τ has rational reduction.

By [HWW22, Proposition 3.8], it extends to a place of L_s of degree one over v. Since L_s contains L, the place v splits in L; splitting at v_0 is automatic. At split places outside B, use points sufficiently close to $E(k_v)$. At inert places outside B, the reduction of τ avoids M, so τ and $p(\tau,1)$ are units; the unramified norm equation (4.4) gives points with both boundary multiplicities zero. These choices map into $\mathcal{N}$ and are Campana on (C_τ,δ_τ) at almost every place, by (4.1). They therefore give a point of the required inverse image. □

Proof of Proposition 4.1. By the reductions above, assume that E is a nonsplit closed point of degree two. Let $\mathcal{N}\subset(Q,\Delta_m)(\mathbf{A}_k)$ be nonempty and open. Lemma 4.4 supplies a smooth conic whose Campana adelic inverse image of $\mathcal{N}$ is nonempty. Lemma 3.1 gives a strict rational point in $\mathcal{N}$, proving density. □

Corollary 4.5. *Let Σ be a smooth quadric surface and B a hyperplane section. Then $(\Sigma,(1-1/m)B)$ satisfies Campana strong approximation for every $m\geq 2$.*

Proof. Apply Proposition 4.1 with $H_1=H_2=B$. □

Remark 4.6. Proposition 4.1 allows coincident sections, common line components, singular sections and tangency between smooth sections. Finiteness of E is needed only for the conic-pencil argument after the initial reductions.

5. Reduction to surfaces

The case $d=2$ was proved in Proposition 4.1. Throughout this section, assume $d\geq 3$ and $E(k)=\varnothing$, the latter by Proposition 3.2. We reduce the finite and positive-dimensional cases to smooth quadric surfaces with two and one hyperplane boundary sections, respectively.

5.1. **Finite common intersection.** First assume that E is finite, hence reduced. Fix

$$\Lambda_0=\Pi_2\cap\cdots\cap\Pi_d\simeq\mathbf{P}^2_k,\qquad B=Q\cap\Lambda_0,\qquad E=B\cap\Pi_1. \tag{5.1}$$

The restriction of the defining quadratic form to the line $\Lambda=\Lambda_0\cap\Pi_1$ has rank two. Hence B is a reduced plane conic, either smooth or a pair of geometric lines. In the latter case its vertex O is k-rational and lies outside H_1, since E is reduced.

For the linear algebra below, write $Q\subset\mathbf{P}(V)$, where V is a k-vector space of dimension $d+2$ and q is its nondegenerate quadratic form. Choose linear forms ℓ_i defining Π_i, and put

$$V_i=\ker(\ell_i),\qquad V_0=\bigcap_{i=2}^{d}V_i,\qquad W_0=V_0\cap V_1.$$

Then $\Lambda_0=\mathbf{P}(V_0)$, $\Pi_1=\mathbf{P}(V_1)$ and $\Lambda=\mathbf{P}(W_0)$, with $\dim V_0=3$ and $\dim W_0=2$. Reducedness of E makes $q|_{W_0}$ nondegenerate.

Lemma 5.1. (1) *There is a nonempty Zariski-open subset $\mathcal{U}\subset Q\setminus D$ such that, for every $R\in\mathcal{U}(k)$,*

$$\Pi_R=\langle\Lambda_0,R\rangle\simeq\mathbf{P}^3_k,\qquad \Sigma_R=Q\cap\Pi_R \tag{5.2}$$

is a smooth quadric surface.

(2) *For every R chosen in* (1)*, the first boundary section $F_1 = H_1|_{\Sigma_R}$ is a reduced plane conic, and the boundary restrictions satisfy*

$$H_i|_{\Sigma_R} = B \quad (2 \le i \le d), \qquad F_1 \cap B = E \tag{5.3}$$

scheme-theoretically. In particular, F_1 and B have no common component.

Proof. (1) The parameter space $\mathcal{P} = \mathbf{P}(V/V_0)$ parametrizes four-dimensional subspaces $U \supset V_0$. The form $q|_{V_0}$ has rank three or two. In the first case, choose $z \in V_0^{\perp}$ with $q(z) \neq 0$; in the second, let o span its radical and choose z with $b_q(o, z) \neq 0$, where b_q is the polar form. In either case $q|_{V_0+kz}$ has rank four. Thus smooth surface sections form a nonempty open subset of $\mathcal{P}$.

Projection from Λ_0 gives

$$Q \setminus \Lambda_0 \longrightarrow \mathcal{P}, \qquad [r] \longmapsto V_0 + kr.$$

It is dominant, since every general smooth surface section contains geometric points outside B. The inverse image of the smooth-section locus under this projection, intersected with $Q \setminus D$, is the required nonempty open subset $\mathcal{U}$.

(2) For $R = [r] \in \mathcal{U}(k)$, put $U_R = V_0 + kr$. For $i \ge 2$, ℓ_i vanishes on V_0 and $\ell_i(r) \neq 0$, because $R \notin D$. Hence $V_i \cap U_R = V_0$, giving $H_i|_{\Sigma_R} = Q \cap \mathbf{P}(V_0) = B$ and $F_1 \cap B = B \cap \Pi_1 = E$ scheme-theoretically. A hyperplane restriction of the rank-four form defining the smooth surface Σ_R has rank at least two, so F_1 is reduced. Finally, finiteness of E excludes a common component of F_1 and B. □

Remark 5.2. The section F_1 may be singular. Its smoothness defines an open condition on $R \in Q \setminus D$. This locus is nonempty by the same nondegeneracy and projection arguments as in the proof of Lemma 5.1, applied to plane sections of $Q \cap \Pi_1$ through Λ. Indeed, $q|_{W_0}$ is nondegenerate and $\operatorname{rank}(q|_{V_1}) \ge d \ge 3$. Denote this nonempty open subset by $\mathcal{U}_1$. Since Q is irreducible, $\mathcal{U} \cap \mathcal{U}_1$ is nonempty and open. Thus the smoothness of F_1 may also be imposed if desired.

Also, reducedness of $E = F_1 \cap B$ forces the vertex of either singular conic to lie outside the other. Both conics are therefore smooth at E and meet transversely there. The optional smoothness of F_1 and the separate transversality assertion are not needed to apply Proposition 4.1.

The following lemma gives the nonempty adelic set needed on the chosen surface.

Lemma 5.3. *Let $\mathcal{N} \subset (Q, \Delta_m)(\mathbf{A}_k)$ be nonempty and open. There exists $R \in \mathcal{U}(k)$ such that, with $\Sigma = \Sigma_R$, $\delta_R = (1 - 1/m)(F_1 + B)$ and $j : \Sigma \hookrightarrow Q$, the set*

$$\mathcal{G} = j_{\mathbf{A}}^{-1}(\mathcal{N}) \subset (\Sigma, \delta_R)(\mathbf{A}_k) \tag{5.4}$$

is nonempty.

Proof. Shrink $\mathcal{N}$ to a basic open subset, taking its finite set T of local conditions large enough to contain the model and reduction exceptions for B and E. For $v \notin T$, choose $b_v \in B(k_v)$ reducing outside H_1: lift a residue-field point outside E when B is smooth, and

otherwise take its vertex. These points satisfy the reference Campana conditions on Q: they lie on H_i for $i \geq 2$ and have zero multiplicity with H_1.

The Hasse principle and weak approximation give $R \in \mathcal{U}(k)$ satisfying the local conditions at T. The tuple equal to R at T and to b_v elsewhere belongs to $(\Sigma, \delta_R)(\mathbf{A}_k)$: at almost every place it lies on B and has zero multiplicity with F_1. Its image lies in $\mathcal{N}$, proving the assertion. □

Remark 5.4 (Geometry of the finite-intersection case)**.** Finiteness of E implies that the hyperplanes are linearly independent and that $\deg E = 2$. Our assumption $E(k) = \varnothing$ makes E reduced, which forces the intermediate plane conic B to be reduced, although it may be singular. A nonreduced E would have the form $2P$ with $P \in Q(k)$, a case already covered by Proposition 3.2; then B may even be a double line. Thus no separate reducedness or smoothness hypothesis is needed in the finite-intersection argument.

5.2. **Positive-dimensional common intersection.** Now assume that $\dim E \geq 1$. With $L = \mathbf{P}(W)$ as in Section 3, the form $q|_W$ is anisotropic over k. Choose a k-plane $L' = \mathbf{P}(V') \subset L$ and put $B = Q \cap L'$. The restriction $q|_{V'}$ is nondegenerate, since a nonzero radical vector would be isotropic. Thus B is a smooth conic.

The rank-three argument in the proof of Lemma 5.1(1) gives a nonempty open subset $\mathcal{U}_+ \subset Q \setminus D$ such that

$$\Sigma_R = Q \cap \langle L', R \rangle \tag{5.5}$$

is a smooth quadric surface for every $R \in \mathcal{U}_+(k)$. Since $R \notin D$ and $L' \subset \Pi_i$, we have

$$H_i|_{\Sigma_R} = B \qquad (1 \leq i \leq d). \tag{5.6}$$

Thus the reduced boundary on Σ_R is the smooth conic B.

Proof of Theorem 1.1. Let $\mathcal{N} \subset (Q, \Delta_m)(\mathbf{A}_k)$ be nonempty and open.

For finite E, Lemma 5.3 gives a smooth surface Σ with boundary $\delta = (1 - 1/m)(F_1 + B)$ and nonempty $\mathcal{G} = j_{\mathbf{A}}^{-1}(\mathcal{N})$. For positive-dimensional E, choose a basic subset of $\mathcal{N}$ with finite set T of local conditions, large enough that $B(k_v) \neq \varnothing$ outside T. The Hasse principle and weak approximation choose $R \in \mathcal{U}_+(k)$ satisfying these local conditions. Put $\Sigma = \Sigma_R$ and $\delta = (1 - 1/m)B$. The tuple equal to R at T and to points of $B(k_v)$ elsewhere belongs to $(\Sigma, \delta)(\mathbf{A}_k)$ and maps into $\mathcal{N}$: outside T, it lies on every boundary component on both Σ and Q. Thus $\mathcal{G}$ is again nonempty.

In both cases $j : (\Sigma, \delta) \hookrightarrow (Q, \Delta_m)$ is an orbifold morphism, so $\mathcal{G}$ is open by Proposition 2.5. Proposition 4.1, respectively Corollary 4.5, gives a strict rational point in $\mathcal{G}$. Its image is strict on Q and belongs to $\mathcal{N}$, proving density. □

References

[Cam11] F. Campana, *Orbifoldes géométriques spéciales et classification biméromorphe des variétés kählériennes compactes*, J. Inst. Math. Jussieu **10** (2011), no. 4, 809–934.

[CT24] J.-L. Colliot-Thélène, *Retour sur l'arithmétique des intersections de deux quadriques*, J. Reine Angew. Math. **806** (2024), 147–185.

[CTSSD87a] J.-L. Colliot-Thélène, J.-J. Sansuc, and P. Swinnerton-Dyer, *Intersections of two quadrics and Châtelet surfaces. I*, J. Reine Angew. Math. **373** (1987), 37–107.

[CTSSD87b] J.-L. Colliot-Thélène, J.-J. Sansuc, and P. Swinnerton-Dyer, *Intersections of two quadrics and Châtelet surfaces. II*, J. Reine Angew. Math. **374** (1987), 72–168.

[CTX09] J.-L. Colliot-Thélène and F. Xu, *Brauer–Manin obstruction for integral points of homogeneous spaces and representation by integral quadratic forms*, Compositio Math. **145** (2009), no. 2, 309–363.

[Har94] D. Harari, *Méthode des fibrations et obstruction de Manin*, Duke Math. J. **75** (1994), no. 1, 221–260.

[Har04] D. Harari, *Weak approximation on algebraic varieties*, in Arithmetic of higher-dimensional algebraic varieties (Palo Alto, CA, 2002), Progr. Math. **226**, Birkhäuser Boston, Boston, MA, 2004, 43–60.

[HWW22] Y. Harpaz, D. Wei, and O. Wittenberg, *Rational points on fibrations with few non-split fibres*, J. Reine Angew. Math. **791** (2022), 89–133.

[HB18] D. R. Heath-Brown, *Zeros of pairs of quadratic forms*, J. Reine Angew. Math. **739** (2018), 41–80.

[MNS24] V. Mitankin, M. Nakahara, and S. Streeter, *Semi-integral Brauer–Manin obstruction and quadric orbifold pairs*, Trans. Amer. Math. Soc. **377** (2024), no. 6, 4435–4480.

[Moe24] B. Moerman, *Generalized Campana points and adelic approximation on toric varieties*, preprint (2024), arXiv:2407.03048.

[NS24] M. Nakahara and S. Streeter, *Weak approximation and the Hilbert property for Campana points*, Michigan Math. J. **74** (2024), no. 2, 227–252.

[SS91] P. Salberger and A. N. Skorobogatov, *Weak approximation for surfaces defined by two quadratic forms*, Duke Math. J. **63** (1991), no. 2, 517–536.

[Wei21] D. Wei, Strong approximation for a toric variety, *Acta Math. Sin. (Engl. Ser.)* **37** (2021), no. 1, 95–103.

[Wei26a] D. Wei, *Campana strong approximation on generalised Markoff surfaces*, preprint (2026), arXiv:2610.13528.

[Wei26b] D. Wei, *Weak approximation for Campana points on projective spaces*, preprint (2026), arXiv:2610.05032.

[WXZ25] D. Wei, J. Xu, and Y. Zhu, *Strong approximation for the intersection of two quadrics*, preprint (2025), arXiv:2412.01144v3.

[Wit07] O. Wittenberg, *Intersections de deux quadriques et pinceaux de courbes de genre 1*, Lecture Notes in Mathematics **1901**, Springer, Berlin, 2007.

Academy of Mathematics and System Science, CAS, Beijing 100190, P. R. China *and* School of Mathematical Sciences, University of CAS, Beijing 100049, P. R. China

Email address: dshwei@amss.ac.cn